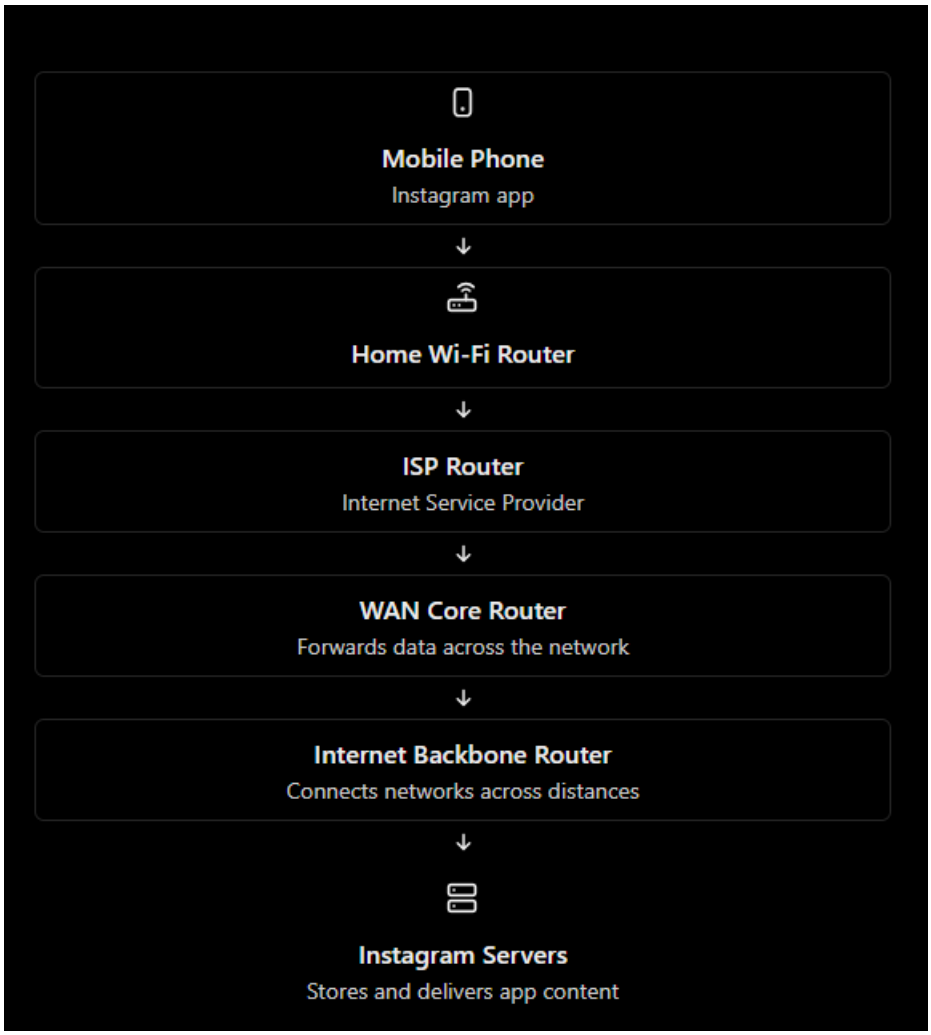
Task 1:

Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Answer:



Explanation: The phone connects to the home Wi-Fi router, which forwards data through the ISP and other internet networks to Instagram's servers. The servers send the requested content back along the network path. The actual route may contain more or different routers.

Task 2.

Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Answer:

WAN Technology	Speed	Cost	Real-World Use
MPLS	High and predictable, depending on the service plan	High	Connecting branches of a large company such as Zomato
Leased Line	Dedicated bandwidth with consistent performance	Very high	Connecting important offices or data centres
Broadband Internet	Medium to very high; performance may vary	Low to medium	Home internet for streaming YouTube or using Instagram

MPLS provides managed connectivity between business locations, leased lines provide dedicated connections, and broadband offers affordable internet access. Actual speeds and prices depend on the provider and service plan.

Task 3.
Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Answer:

A VPN (Virtual Private Network) creates an encrypted tunnel over a public network such as the internet. It allows a user to communicate more securely when connected to public Wi-Fi at a coffee shop.

How it works:

1. The user connects their phone or laptop to the coffee shop's Wi-Fi.
2. The VPN app establishes an encrypted connection with a VPN server over the WAN.
3. Data travels through the encrypted tunnel, making it harder for people on the public network to read or modify it.
4. The VPN server forwards traffic to services such as Spotify or YouTube and returns the responses through the VPN connection.

Example: If you watch YouTube while connected to a VPN, your traffic between your device and the VPN server is encrypted. A VPN does not guarantee complete anonymity or protect against every threat, so HTTPS and safe browsing practices remain important.

Task 4:

Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Answer:

I recommend MPLS (Multiprotocol Label Switching) to connect Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi.

Two reasons:

1. Reliable connectivity: MPLS can provide predictable network performance for warehouse inventory systems, order processing, and communication between locations.
2. Quality of Service (QoS): It can prioritize important business traffic, helping critical applications operate consistently across the three warehouses.

MPLS is a strong choice when reliability and predictable performance are priorities, although it usually costs more than broadband. Flipkart could also consider broadband with site-to-site VPN or SD-WAN if reducing costs and adding flexible backup links are more important.